

Welcome to the Caltrans Utility AI Assistant

<<

Caltrans Context

Select Caltrans District:

?

District 11



Caltrans Utility AI Assistant

Choose Assistant Mode:

☒  Standalone Manual Lookup incl SOP (Text)

☐  Automated Field Photo Scanner (Image)

☐

 Activate Section 77 Special Provisions Drafting Wizard



Query Caltrans & SOP Manuals Directly

Type your engineering or utility question here...

For a third-party Encroachment Permit, is the permittee required to prepare and submit a Utility Management Matrix (UMM)? Use only the supplied refe



Manual Lookup Profile

Ready to search standard SOP database using District 11 clearance baselines.

 Search Manuals & SOPs



Engineering Directive

Executive Summary: Caltrans Utility AI Assistant

The **Caltrans Utility AI Assistant** is an engineering-focused digital system that connects technical reference material, project-developed engineering knowledge, and field-oriented workflows in a single assistant.

Current Capabilities

- **Retrieval-Augmented Engineering Assistance** — Retrieves relevant engineering references at runtime and provides source-grounded answers with source and page references where available.
- **Engineering Knowledge Integration** — Combines formal references with project-developed Engineering Notes, procedures, supplemental guidance, and utility-owner-specific knowledge.
- **Automated Specification Drafting** — Assists with development of utility-related specification language using retrieved references and project guidance.
- **Intelligent Object Detection** — Uses computer vision to identify transportation and utility infrastructure from photographs and connect detected assets with relevant engineering guidance.
- **Workflow-Aware Assistance** — **Incorporates District, utility-owner, project, and workflow** context when evaluating engineering questions.

Planned Expansion

The system is being developed toward a broader **multimodal and geospatial engineering assistant**, including:

- **Conversational Engineering Support** — Identify missing project information, ask targeted follow-up questions, and continue an engineering evaluation interactively.
- **GIS-Aware Engineering Analysis** — Incorporate spatial and project-location information into retrieval and engineering evaluation.
- **Photo + RAG Analysis** — Combine photographs, detected infrastructure, project context, and engineering references in a single workflow.
- **Pavement and Roadway Condition Detection** — Extend computer vision to pavement defects, utility cuts, patches, and other roadway conditions.
- **Advanced Engineering Workflow Orchestration** — Connect document retrieval, computer vision, GIS, and conversational reasoning into coordinated engineering workflows.

Impact

The objective is not to replace engineering judgment, but to **reduce repetitive work, improve access to specialized engineering knowledge, support field identification, and make technical requirements more traceable and accessible.**

The architecture is intentionally modular, allowing models, data sources, and specialized AI capabilities to evolve without rebuilding the underlying engineering knowledge and retrieval system.

 **Caltrans Context**

Select Caltrans District: 

District 11 

OPTION:
CHOOSE YOUR DISTRICT

A:
LOOKUP MANUAL
*ASK ANYTHING

D:
GIS & REFERENCE DOCS

Caltrans Utility AI Assistant

Choose Assistant Mode:

A:



 Standalone Manual Lookup incl SOP (Text)

B:



 Automated Field Photo Scanner (Image)

C:



 Activate Section 77 Special Provisions Drafting Wizard

4x OPTIONS:
(desktop or
Cell Ph)

- A. MANUAL LOOKUP
- B. SCAN A FIELD UTILITY
OR UPLOADED PHOTO
- C. SPEC S77 DESIGN
- D. GIS & REFERENCE DOCS
(below)



Query Caltrans & SOP Manuals Directly

Type your engineering or utility question here...

e.g., What are the clearance requirements in District 11?



 GIS Reference Map

Note: All tools are available on Desktop or on Cell Phone
SOP – Standard Operating Procedure

C. SPEC S77 DESIGN

Caltrans Context

Select Caltrans District:

District 11



Caltrans Utility AI Assistant

Choose Assistant Mode:



Standalone Manual Lookup incl SOP (Text)



Automated Field Photo Scanner (Image)



Activate Section 77 Special Provisions Drafting Wizard



Section 77 Special Provisions Drafting Wizard

Confirm Specific Target Asset:

Water Main



Select Operating Utility Agency/Owner:

San Diego County Water Authority



Select Material Base:

PVC (C900 / C905)



Dimension / Diameter Class:

N/A



Operational Scope of Work:

Install / Construct New System



Engineering Overrides & Quantity Inputs:

Primary Bid Item Payment Unit:

Linear Foot (LF)



Enter Proposed Quantity (matching Linear Foot (LF)):

10.00



☐ Utility Crossing Requires Outer Protective Casing / Sleeve

☐ System Carries Recycled / Purple Pipe Non-Potable Water

Drafting contract framework using District 11 parameters.

B. SCAN A FIELD UTILITY OR UPLOADED PIC

Caltrans Context

Select Caltrans District:

District 11

Data Input

YOLO Confidence Threshold

0.35

CHOOSE THRESHOLD

Caltrans Utility AI Assistant

Choose Assistant Mode:

Standalone Manual Lookup incl SOP (Text)

Automated Field Photo Scanner (Image)

Select Image Input Method:

Upload Image File

Use Live Device Camera

Upload a Street/Field Image...

Upload

200MB per file • JPG, PNG

Upload a field image in the sidebar to begin asset verification.

>

GIS Reference Map

>

Engineering Exhibits & Reference Materials

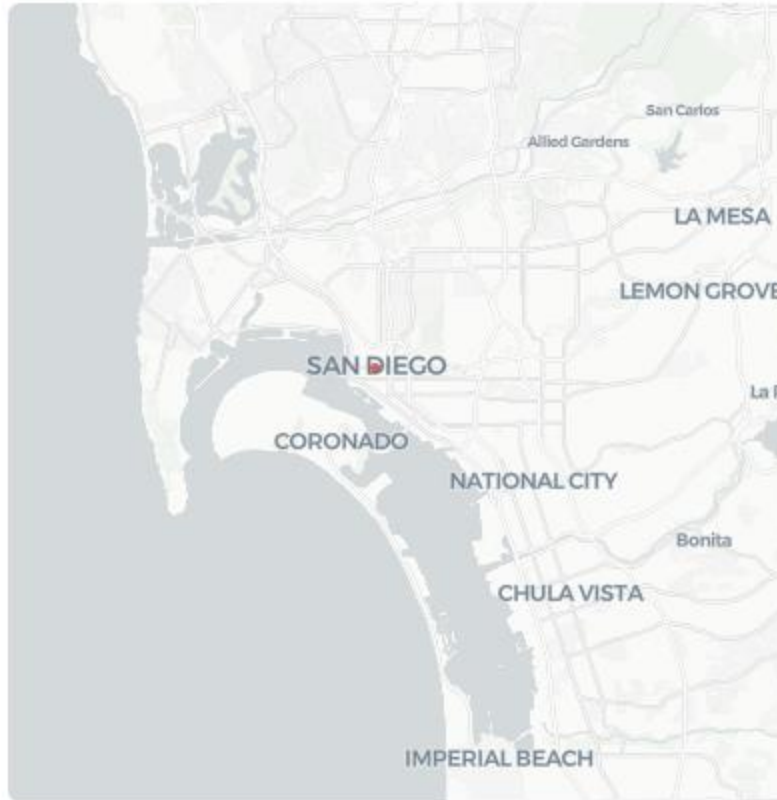
METHOD OF CHOICE
TO LOAD ITEM

D. GIS & REFERENCE DOCS

GIS Reference Map

 Mapbox GL Project Alignment Engine

Displaying regional map layouts centered over the San Diego District 11 corridor.

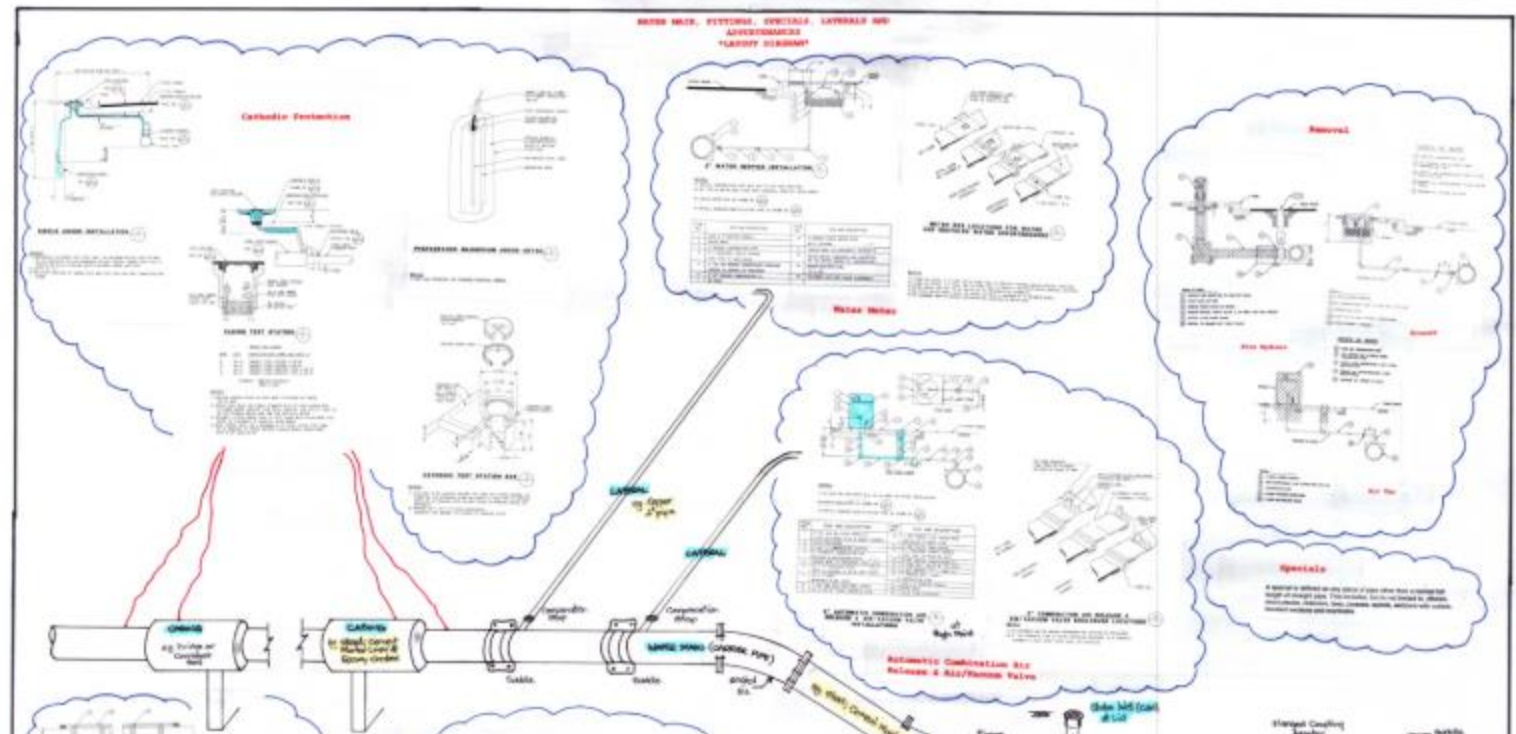


 San Diego Manning Interface Active: Canvas is clear, stable, and re

- Engineering Exhibits & Reference Materials

 Caltrans Standard Water Main Assemblies

Pro-Tip: Right-click the image and select 'Open image in new tab' to view, zoom, and inspect blueprint details in full resolution.



A. MANUAL LOOKUP



Caltrans Utility AI Assistant

Choose Assistant Mode:

☒ Standalone Manual Lookup incl SOP (Text) ☐ Automated Field Photo Scanner (Image)

☐ Activate Section 77 Special Provisions Drafting Wizard



Query Caltrans & SOP Manuals Directly

Type your engineering or utility question here...

What is a UMM, and must I do one?

Question: What is a UMM, and must I do one?

The UMM is discussed as needed as part of the RTL process.

Under Additional Information the suggestions are whether you are a Caltrans representative or a Permittee/permit application.

You might follow up with new revised questions (result shown):

Engineering Finding

No. Based on the supplied Encroachment Permits Manual and the District 11 utility engineering SOPs, a third-party permittee is **not required to prepare a Utility Management Matrix (UMM)** as part of an encroachment permit application package.

Engineering Finding

Yes, you must maintain a Utility Management Matrix (UMM) while preparing PS&E utility engineering for Caltrans.

Key Requirements

- PDPM, p. 35 – “Ultimately, information from the UMM will be entered in the CUD. ... Regardless of the project sponsor, the district UEW (as applicable) is the recipient of the UMM.”

Engineering Finding

A UMM is the **Utility Management Matrix** – a tracking document used to record utility ownership, location, conflict status, clearance determinations, and related data that is ultimately entered into the Caltrans Utility Database (CUD).

Key Requirements

- The Project Development Procedures Manual (PDPM) states that utility location data is used to “populate the UMM” and that, regardless of project sponsor, the district Utility Engineering Workgroup (UEW) is the recipient of the UMM, with a copy provided to the district right-of-way utility coordinator **【PDPM p. 35, Locating Requirements Overview】** .
- The PDPM also notes that the PE, with input from the UEW, must practice due diligence to collect utility information and “populate the UMM” **【PDPM p. 34, Policies】** .
- SOP-UE-013 (Phase 1 PS&E Utility Engineering Activities) lists “Maintain the Utility Management Matrix (UMM)” as a responsibility of the Nominated Utility Engineer during the PS&E phase.
- SOP-UE-014 (Maintenance Emergency, Minor B, DO, and DDO Utility Review) includes the production of a “UMM Utility Management Matrix: List all utilities and their clearance status” as part of the deliverable package for Minor B and similar time-sensitive projects.

Note: this blue text is the AI Assistant produced output

Conditions / Exceptions

- The requirement to prepare and maintain a UMM applies to **Caltrans-administered projects** (including PS&E, Minor B, Director’s Order, Deputy Director’s Order, and other utility-engineering workflows described in the PDPM and the SOPs).
- The supplied documents do **not** impose a UMM obligation on third-party encroachment-permit applicants unless the specific permit’s special provisions, agreement, or project-specific direction explicitly requires it. No such third-party requirement appears in the provided context.
- If you are a utility engineer working on a Caltrans-led project (PS&E, Minor B, etc.), you must prepare, maintain, and submit the UMM as described. If you are a third-party permit applicant, the UMM is not automatically required by the cited references.

Additional Information Needed

To confirm whether a UMM is mandatory for your specific situation, please clarify:

1. **Your role** – Are you a Caltrans utility engineer (or consultant working for Caltrans) or a third-party encroachment-permit applicant?
2. **Project type** – Is the work a PS&E effort, a Minor B/Maintenance Emergency project, or another Caltrans-administered effort?
3. **Any permit special provisions** – If you are a third-party applicant, does the encroachment permit or associated agreement explicitly call for a Utility Management Matrix?

Providing these details will allow a definitive answer on whether you must prepare a UMM.

Engineer Verification

Not required once the above role and project-type information is supplied; the UMM requirement is explicitly stated in the PDPM and the referenced SOPs for Caltrans-administered utility engineering work.

A. MANUAL LOOKUP Continued



Caltrans Utility AI Assistant

Choose Assistant Mode:

☒  Standalone Manual Lookup incl SOP (Text) ☐  Automated Field Photo Scanner (Image)

☐  Activate Section 77 Special Provisions Drafting Wizard

Query Caltrans & SOP Manuals Directly

Type your engineering or utility question here...

I'm a utility engineer working on a Emergency Maintenance project, do I have to prepare plans and sign a Cert?

Engineering Finding

Yes, you must sign a Utility Certification, and you must prepare utility-related plans/documentation. However, for an Emergency Maintenance project the documentation package is not the full PS&E plan set; it is a streamlined emergency deliverable set.

Key Requirements

- **Utility Certification:** The Utility Policy Certification is mandatory. It must be signed by a California registered civil engineer and attached to the project submittal (PDPM p. 47).
- **Emergency Deliverables:** For Maintenance Emergency, Minor B, Director's Order, or Deputy Director's Order projects, the utility engineer is required to produce a **Minor B Checklist**, a **Utility Management Matrix (UMM)**, layout sketches (e.g., a .DGN with raster overlay), and the signed Utility Certification (SOP-UE-014 §3.4, §3.5).
- **Plan Preparation:** The "plans" you must prepare are the utility sketches and the UMM that show proposed work limits versus known utilities, not the standard contract plan sheets (SOP-UE-014 §3.3, §3.4).

Conditions / Exceptions

- The streamlined requirements of SOP-UE-014 apply only when the project is classified as a Maintenance Emergency, Minor B, DO, or DDO. If the project does not fall into one of these categories, the standard PDPM utility plan and certification rules apply.
- If a policy deviation is needed, a Utility Policy Exception (UPE) must be obtained and attached to the certification (PDPM p. 47; SOP-UE-014 §3.3).

Additional Information Needed

To confirm the exact deliverables, the specific project classification (e.g., Minor B, DO) and the scope of utility conflicts should be provided.

Engineer Verification

If the project classification is unclear or the scope involves high-priority utilities that may trigger additional field verification, consult the district utility lead to verify the applicable workflow.

B. SCAN A FIELD UTILITY OR UPLOADED PIC (Computer Vision)



Note: change threshold to discover more items

For an electric transformer associated with SDG&E in Caltrans District 11, the applicable Caltrans requirements include encroachment permitting, specific installation standards, clearance requirements, location protocols, and protection measures. Electric power lines and high-voltage electric supply lines are classified as high-priority utilities under California Government Code § 4216, which mandates specific Caltrans procedures for their handling.

Note: AI Assistant Generated

Key Requirements

Encroachment Permit An encroachment permit is required for the installation of new utilities or relocation of utilities within the State highway right-of-way (Encroachment Permits Manual, Page 33). The project must obtain this permit before proceeding with construction.

Installation Standards Utilities must meet minimum cover and clearance requirements of 42 inches below finished grade or 18 inches below the grading plane (Project Development Procedures Manual, Page 39). If the transformer requires relocation within the right-of-way, it must meet requirements for new installations (Encroachment Permits Manual, Page 33).

Upload a Street/Field Image...



street_...ilities.png
2.8MB



Automated Asset Detection



Caltrans Manual & Owner Asset Intelligence

Detected 8 asset types!

Select an asset type to view Caltrans Guidelines:

water_fire_hydrant

Identify Asset Owner / Franchise Holder (Optional context for proprietary rules):

General / Unknown Owner

> Plans Preparation Manual (PPM) Drafting Guidelines

Search Manuals & Generate Directive

B. SCAN A FIELD UTILITY OR UPLOADED PIC (Computer Vision) Continued



Automated Asset Detection



Caltrans Manual & Owner Asset Intelligence

Detected 1 asset types!

Select an asset type to view Caltrans Guidelines:

telecom_mh_lid

Identify Asset Owner / Franchise Holder (Optional context for proprietary rules):

AT&T California

Plans Preparation Manual (PPM) Drafting Guidelines

1 Cross-referenced from Caltrans PPM Section 2-2.18: Utility Sheets (U-Sheets)

- **U-Sheet Linework:** Existing `telecom_mh_lid` facilities must be drawn in a screen/dashed light weight. Proposed utility installations or modifications must be drawn in a solid, bold black line weight.

• **Asset Labels:** Clearly label the asset owner name, franchise agreement number,

1. Permit Classification

Telecommunications and wired broadband facilities are classified under **Section Code 603.2A** (Encroachment Permits Manual, Table 6.0B, Page 4). A valid encroachment permit is **required** before any installation within State right-of-way.

2. Concurrence Requirement for Shared Conduit

If AT&T California is proposing to place new telecommunication cabling into an existing facility that falls under "prior rights" or an "exception to policy" belonging to another telecommunications franchise, AT&T **must submit proof of concurrence** from the owning telecommunications franchise by means of an agreement, letter, or contract when submitting the encroachment permit application (Encroachment Permits Manual, Section 603.2A-2, Page 38).

3. Minimum Cover for Subsurface Installations

Note: AI Assistant Generated

C. SPEC S77 DESIGN

 Caltrans Utility AI Assistant

Choose Assistant Mode:

☒  Standalone Manual Lookup incl SOP (Text) ☐  Automated Field Photo Scanner (Image)

☒  Activate Section 77 Special Provisions Drafting Wizard

 Section 77 Special Provisions Drafting Wizard

Confirm Specific Target Asset:

Sewer Manhole

Select Operating Utility Agency/Owner:

City of San Diego Sewer Department

Select Material Base:

N/A - Structural Adjustment ...

Dimension / Diameter Class:

N/A

Operational Scope of Work:

Adjust Existing Facility to Gr...

 Engineering Overrides & Quantity Inputs:

Primary Bid Item Payment Unit:

Each (EA)

Enter Proposed Quantity (matching Each (EA)):

3.00

☐ Utility Crossing Requires Outer Protective Casing / Sleeve

☐ System Carries Recycled / Purple Pipe Non-Potable Water

Drafting contract framework using District 11 parameters.

 **Generate Comprehensive Contract Draft & Estimate**

Note: AI Assistant Generated

Unit Rate	Quantity Input	10% Mobilization & Buffer	Total Proposed Budget
\$4,500.00	3.0 (EA)	\$1,350.00	\$14,850.00

 Generated Section 77 Contract Draft

Engineering Finding

The proposed work falls under **Scope Tier 3: Sewer Manhole Adjustment (Raise or Lower)** per the SSP-S77 boilerplate template. The work consists of adjusting an existing sewer manhole frame and cover to conform to the proposed finished grade. Protective casing/sleeve is explicitly **NOT REQUIRED**, consistent with the Encroachment Permits Manual §603.3C exception for gravity flow sewer lines. The applicable governing references are the SSP-S77 template, PDPM Chapter 17 (Encroachments and Utilities), and the Encroachment Permits Manual Chapter 600.

SECTION 77-XX — SEWER MANHOLE ADJUSTMENT TO GRADE

PART I — GENERAL

77-XX.01 Description

The work covered by this section consists of furnishing labor, equipment, materials, coordination, and incidentals

PART I — GENERAL

C. SPEC S77 DESIGN Cont.

77-XX.01 Description

The work covered by this section consists of furnishing labor, equipment, materials, coordination, and incidentals necessary to adjust an existing sewer manhole frame and cover to the proposed finished grade shown on the approved contract plans.

Work includes:

- Exposing the existing manhole structure.
- Removing existing grade adjustment mat
- Establishing the proposed final elevation.
- Installing approved adjustment rings, ext

77-XX.03 Utility Owner Coordination

The Contractor shall coordinate with the affected Utility Owner before beginning utility work. Coordination may include advance notification, inspection requirements, material approval, testing, existing facility verification, and acceptance of completed work.

77-XX.04 Submittals

When required by the contract documents or Utility data, material certifications, installation details, an periods established by the contract documents and the affected work begins.

77-XX.08 Concrete

Concrete for collars, grade rings, or surface treatment shall conform to the applicable Caltrans Standard Specifications for concrete. Strength, material type, dimensions, and placement requirements shall be verified against the approved plans and Utility Owner requirements.

[VERIFY APPLICABLE CALTRANS STANDARD SPECIFICATION SECTION FOR CONCRETE REQUIREMENTS]

Note: AI Assistant Generated

PART II — MATERIALS

77-XX.05 General

Materials shall conform to the applicable Caltrans Plans, and Utility Owner requirements. The applica where required.

77-XX.06 Adjustment Materials

PART III — CONSTRUCTION

77-XX.09 General Construction Requirements

Construct the sewer manhole adjustment in accordance with the approved Utility Plans, project specifications, Utility Owner requirements, and applicable Caltrans Standard Specifications.

Before beginning work, the Contractor shall verify:

- Existing utility locations.
- Proposed utility alignment.
- Connection points.
- Existing and proposed elevations.

77-XX.18 Measurement

Measure each completed sewer manhole adjustment as specified in the contract. Measurement shall be by each completed adjustment (EA).

77-XX.19 Payment

Payment for each completed adjustment includes:

- Applicable adjustment materials (adjustment rings, grade rings)
- Frame and cover adjustment and resetting.
- Concrete collar or surface treatment.
- Excavation, exposure, and backfill.
- Surface restoration.
- Testing and inspection as required.
- Utility Owner acceptance.

CONDITIONS / EXCEPTIONS ⇄

1. **Encasement Exemption:** Protective casing/sleeve is NOT REQUIRED for this work. This is consistent with Encroachment Permits Manual §603.3C, which exempts gravity flow sewer lines from the encasement requirement.
2. **Clearance Compliance:** The adjusted facility must still meet PDPM Chapter 17, Article 4 clearance and offset requirements, including depth of cover.
3. **Utility Owner Requirements:** City of San Diego Sewer Department-specific technical standards for manhole adjustment were not available in the supplied reference materials. The Contractor must comply with the applicable Utility Owner standards as identified in the approved Utility Plans and U-Sheets. [VERIFY CITY OF SAN DIEGO SEWER DEPARTMENT MANHOLE ADJUSTMENT STANDARDS]
4. **Cal-OSHA / Government Code 4216:** If the adjustment involves any milling, grinding, or cold planing of adjacent pavement, or if excavation occurs near high-priority utilities, additional statutory safety requirements apply regardless

ADDITIONAL INFORMATION NEEDED

The following information would allow further evaluation and refinement of the provisions:

- **Proposed finished grade elevation** and whether the adjustment involves raising or lowering the manhole
- **Existing manhole condition** (frame type, cover type, material).
- **Specific City of San Diego Sewer Department technical standards** for manhole adjustment (materials, n acceptance criteria).
- **Applicable Caltrans Standard Specification section numbers** for concrete, adjustment materials, and su

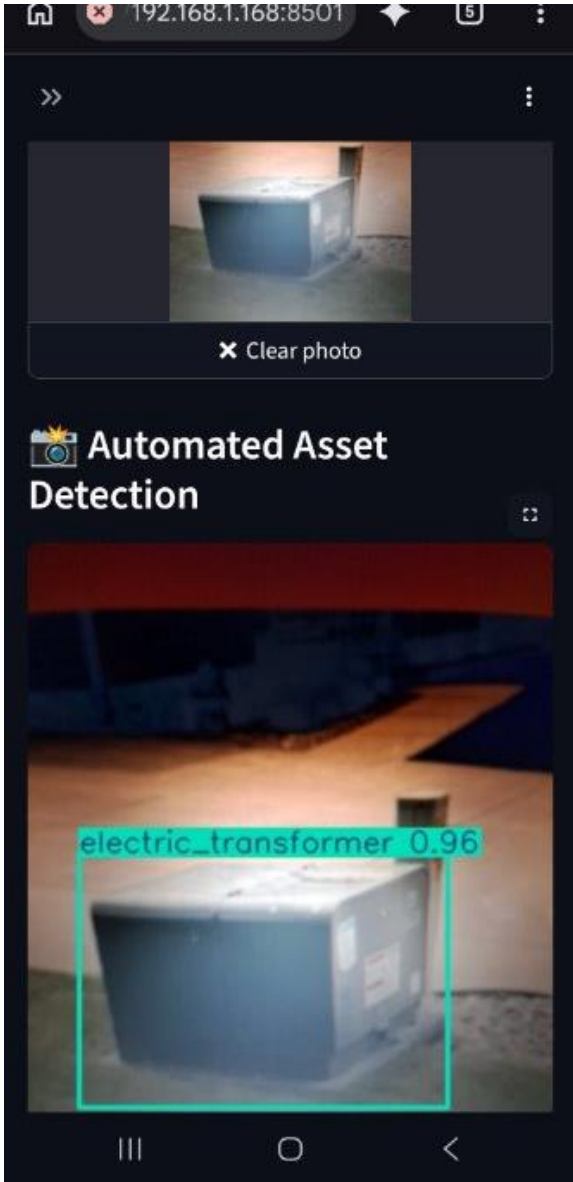
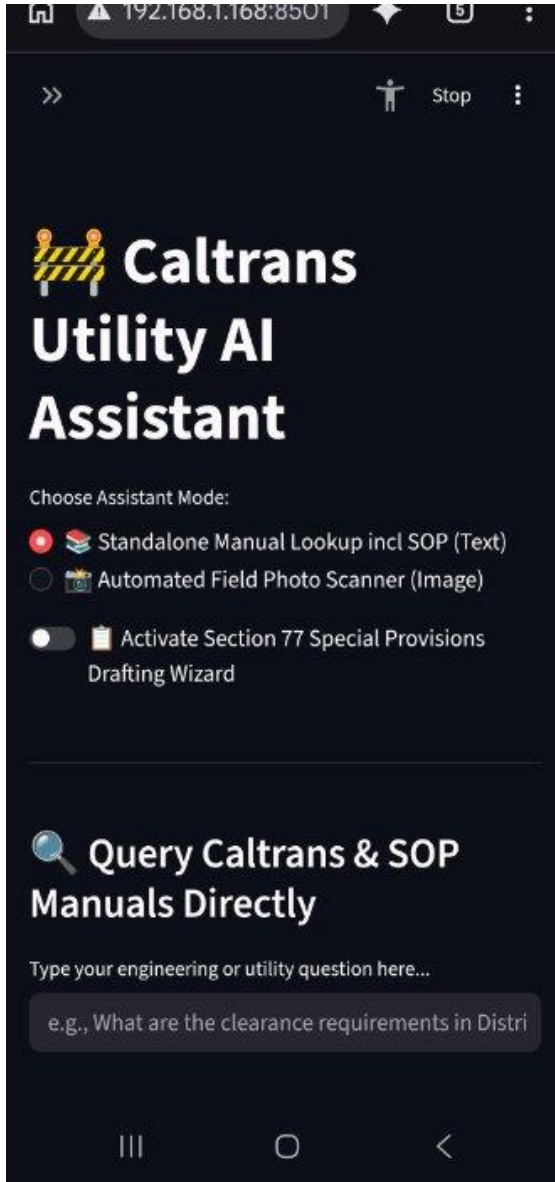
Note: AI Assistant Generated

ENGINEER VERIFICATION

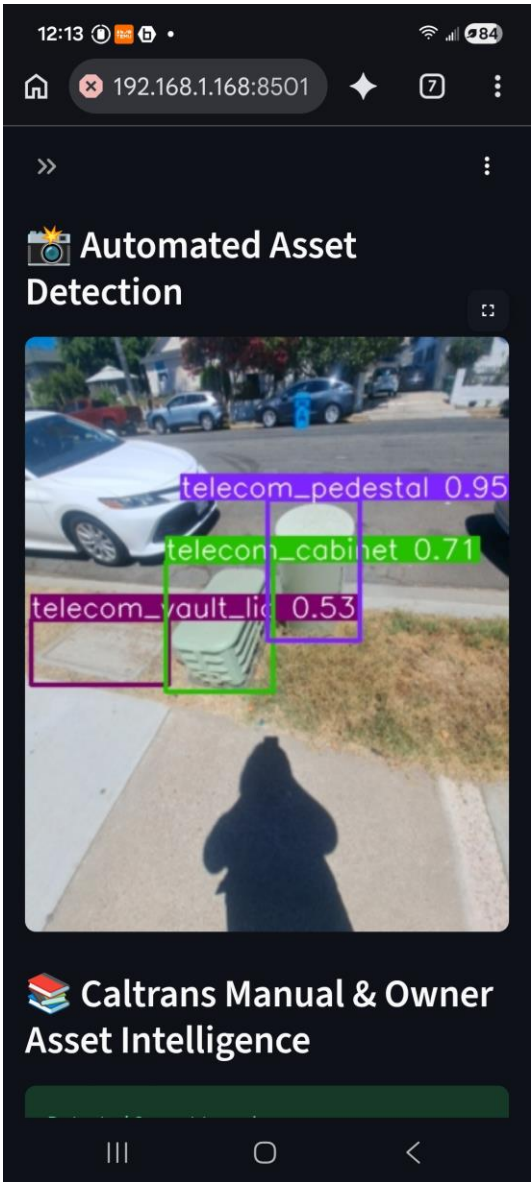
The following items require project-specific engineer verification and cannot be resolved from the supplied references alone:

1. **City of San Diego Sewer Department manhole adjustment standards** — No owner-specific technical document for sewer manhole adjustment was available in the reference database. The applicable Utility Owner standards must be verified against the approved Utility Plans and U-Sheets.
2. **Specific Caltrans Standard Specification sections** — The exact Caltrans Standard Specification sections governing concrete strength, adjustment materials, and surface restoration for this work were not cited in the supplied context. Placeholders have been included for engineer completion.
3. **Right-of-Way Master Agreement applicability** — Whether any active District Right-of-Way Master Agreement affects

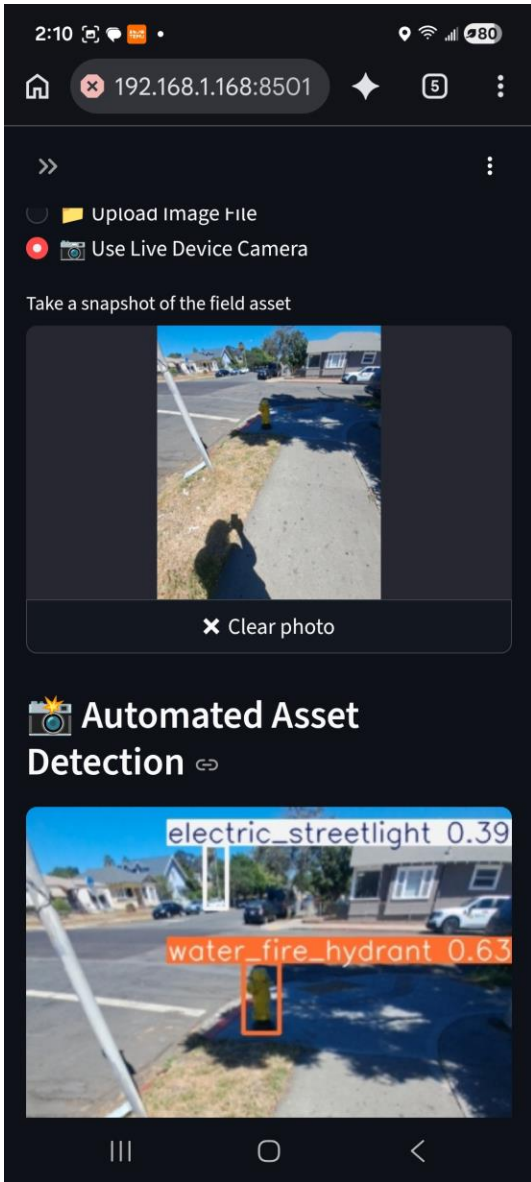
CELL PH APPLICATION (as before, now on Cell phone)



UPLOAD PIC



STREET LIVE PIC



Where It Goes Next

The current assistant establishes the foundation. Future development can extend that foundation from answering questions to supporting the broader engineering workflow.

Conversational Follow-Up

- The assistant identifies missing project information.
- Instead of requiring the user to start over, it asks the next relevant question and continues the analysis.

Photo + RAG

- Connect computer-vision detection with the engineering knowledge base.
- A detected asset can trigger retrieval of relevant requirements, guidance, or engineering notes.

GIS-Aware Analysis

- Incorporate location and spatial context into engineering questions and asset identification.
- Connect detected or identified utilities with their geographic/project context.

Pavement & Roadway Detection

- Extend computer vision beyond utility assets to pavement and roadway conditions.
- Creates a potential path toward broader field-condition assessment.

Workflow Orchestration

- Move beyond individual answers toward assistance with multi-step engineering workflows, documentation, reviews, and deliverables.

The direction is not simply “more AI.” It is deeper integration between engineering knowledge, project context, field information, and the work engineers actually perform.

Why This Is Different

A general-purpose AI model can generate an answer.

This assistant is designed to work within an engineering knowledge environment.

1. Grounded in Engineering References

The assistant retrieves from the specific manuals, standards, procedures, and project references relevant to the question.

2. Incorporates Organizational Knowledge

Project-developed Engineering Notes, SOPs, historical information, and owner-specific knowledge can be incorporated into the searchable knowledge base.

This is where the system becomes more than an internet search or general-purpose AI response.

3. Understands Context

The response can be influenced by factors such as:

Project Path • District • Utility Type • Utility Owner • Location • Construction Activity

rather than treating every question as an isolated question.

4. Designed for the Workflow

The objective isn't simply to produce an answer. The assistant can support:

Research → Review → Engineering Direction → Documentation → Field Support

The underlying AI model is replaceable. The valuable asset is the engineering knowledge, workflow logic, context, and application built around it.